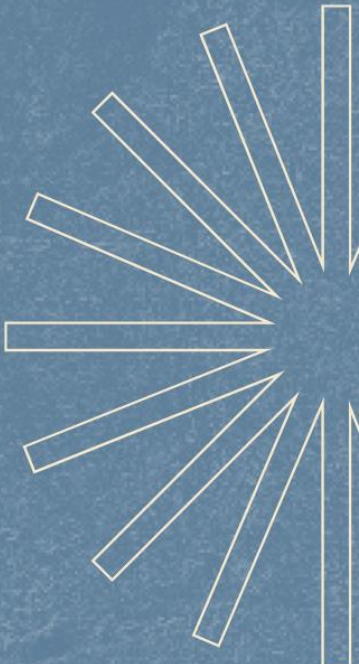
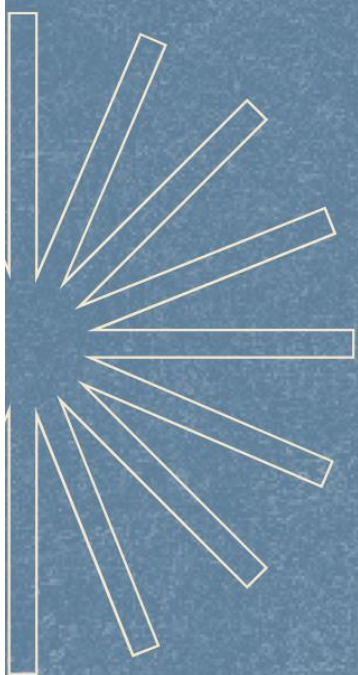




DSA0602

DATA HANDLING & VISUALIZATION

Lab Experiments



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Set 14. Dataset: Survey Results

R PROGRAM:

```
# Survey Results Dataset
respondent <- c(1,2,3)
q1 <- c("A","B","C")
q2 <- c("B","A","A")
q3 <- c("C","D","B")

# Convert responses into numeric values for radar chart
q1_score <- c(1,2,3)
q2_score <- c(2,1,1)
q3_score <- c(3,4,2)

# Display 3 outputs together
par(mfrow=c(1,3))

# 1. Stacked Bar Chart
response <- rbind(
  table(factor(q1, levels=c("A","B","C","D"))),
  table(factor(q2, levels=c("A","B","C","D"))),
  table(factor(q3, levels=c("A","B","C","D")))
)

barplot(response,
  beside=FALSE,
  col=c("red","green","blue"),
  main="Survey Response Distribution",
  xlab="Answers",
  ylab="Frequency",
  legend.text=c("Question 1","Question 2","Question 3"))

# 2. Radar Chart
library(fmsb)

radar_data <- data.frame(
  Q1=q1_score,
  Q2=q2_score,
  Q3=q3_score
)

# Add maximum and minimum values
radar_data <- rbind(
  c(4,4,4),
  c(0,0,0),
  radar_data
)

radarchart(radar_data,
  axistype=1,
  pcol=c("red","blue","green"),
  pfcol=c(rgb(1,0,0,0.3),
    rgb(0,0,1,0.3),
    rgb(0,1,0,0.3)),
  plwd=2,
```

```

        title="Survey Response Pattern")

# 3. Table
plot.new()
title("Survey Response Table")

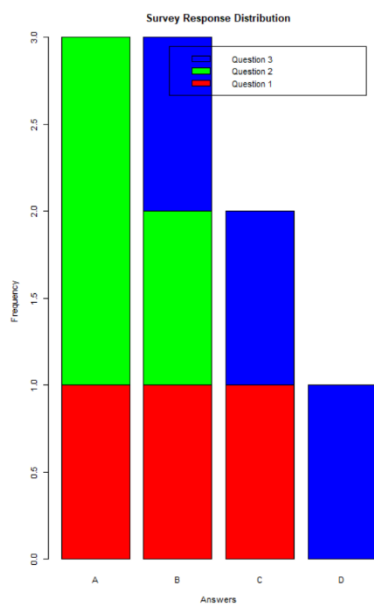
table_data <- data.frame(
  Respondent=respondent,
  Question1=q1,
  Question2=q2,
  Question3=q3
)

text(0.5,0.8,
     paste(capture.output(print(table_data)),
           collapse="\n"),
     family="mono")

# Reset Layout
par(mfrow=c(1,1))

```

OUTPUT:

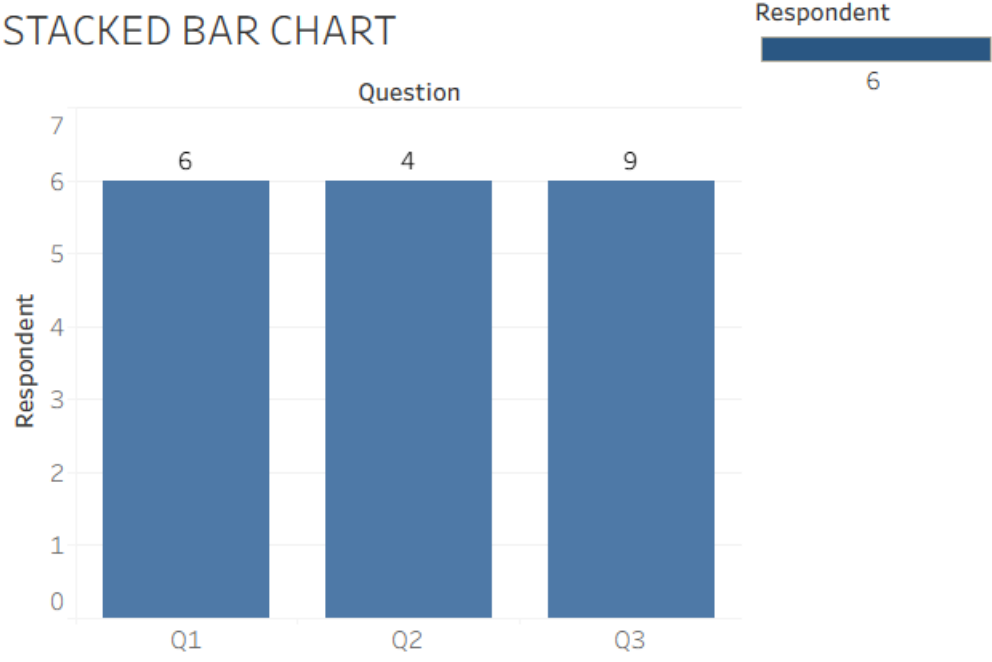


Survey Response Table

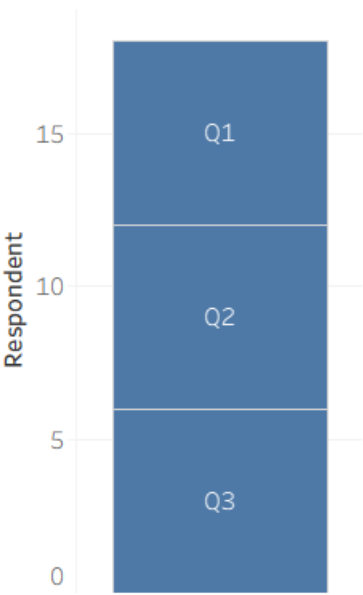
Respondent	Question1	Question2	Question3
1	A	B	C
2	B	A	D
3	C	A	B

TABLEAU DASHBOARD:

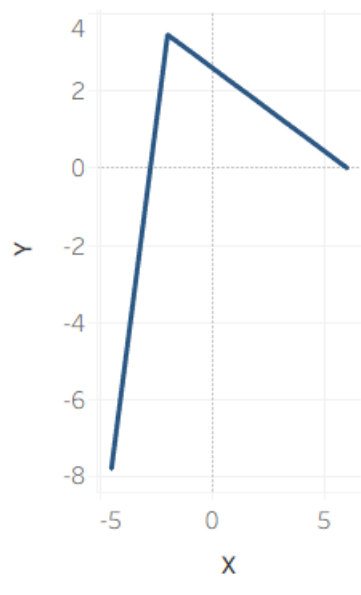
STACKED BAR CHART



SURVEY TABLE



Sheet 3



Set 15: Product Sales Analysis

R PROGRAM:

```
# Monthly Product Sales Dataset
product <- c("Product A","Product B","Product C")
jan <- c(2000,1500,1200)
feb <- c(2200,1800,1400)
mar <- c(2400,1600,1100)
```

```
# Display 3 outputs together
par(mfrow=c(1,3))
```

```
# 1. Grouped Bar Chart
sales <- rbind(jan,feb,mar)
```

```
barplot(sales,
        beside=TRUE,
        col=c("red","green","blue"),
        names.arg=product,
        main="Quarter 1 Product Sales",
        xlab="Product",
        ylab="Sales",
        legend.text=c("January","February","March"))
```

```
# 2. Stacked Area Chart
month <- c(1,2,3)
```

```
plot(month, jan,
      type="n",
      xaxt="n",
      ylim=c(0,max(jan+feb+mar)),
      xlab="Month",
      ylab="Sales",
      main="Overall Sales Trend")
```

```
axis(1,
     at=1:3,
     labels=c("January","February","March"))
```

```
polygon(c(month,rev(month)),
        c(rep(0,3),rev(jan)),
        col="lightblue",
        border=NA)
```

```
polygon(c(month,rev(month)),
        c(jan,rev(jan+feb)),
        col="lightgreen",
        border=NA)
```

```
polygon(c(month,rev(month)),
        c(jan+feb,rev(jan+feb+mar)),
        col="pink",
        border=NA)
```

```
legend("topleft",
```

```

        legend=product,
        fill=c("lightblue","lightgreen","pink"))

# 3. Table
plot.new()
title("Monthly Product Sales Table")

table_data <- data.frame(
  Product=product,
  January=jan,
  February=feb,
  March=mar
)

text(0.5,0.8,
     paste(capture.output(print(table_data)),
           collapse="\n"),
     family="mono")

# Reset Layout
par(mfrow=c(1,1))

```

OUTPUT:

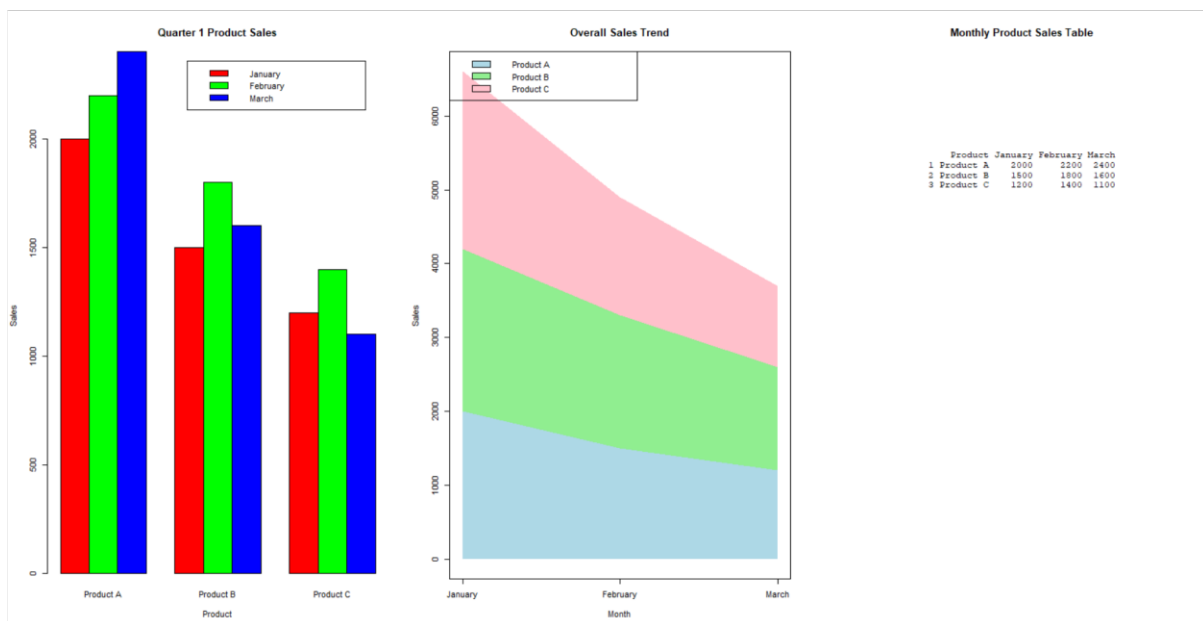
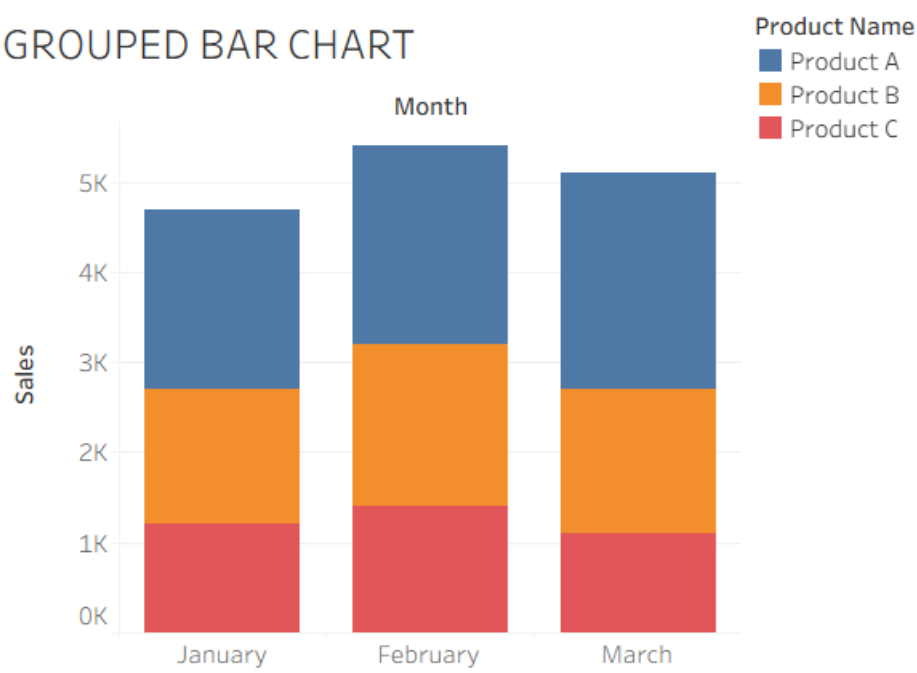
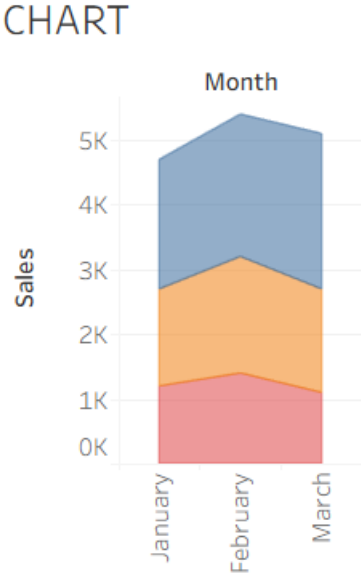


TABLEAU DASHBOARD:

GROUPED BAR CHART



STACKED AREA CHART



Sheet 3

Product ..	Month		
	Janua..	Febru..	Mar
Product A	2,000	2,200	2,
Product B	1,500	1,800	1,
Product C	1,200	1,400	1,

Set 16: Customer Demographics Analysis

R PROGRAM:

```
# Customer Demographics Dataset
customer_id <- c(1,2,3)
age <- c(28,35,42)
gender <- c("Female","Male","Female")
income <- c(50000,60000,75000)

# Display 3 outputs together
par(mfrow=c(1,3))

# 1. Bar Chart
barplot(age,
        names.arg=customer_id,
        col="skyblue",
        main="Customer Age Distribution",
        xlab="Customer ID",
        ylab="Age")

# 2. Pie Chart
pie(table(gender),
    col=c("pink","lightblue"),
    main="Customer Distribution by Gender")

# 3. Table
plot.new()
title("Customer Demographics Table")

table_data <- data.frame(
  CustomerID=customer_id,
  Age=age,
  Gender=gender,
  Income=income
)

text(0.5,0.8,
     paste(capture.output(print(table_data)),
           collapse="\n"),
     family="mono")

# Reset Layout
par(mfrow=c(1,1))
```

OUTPUT:

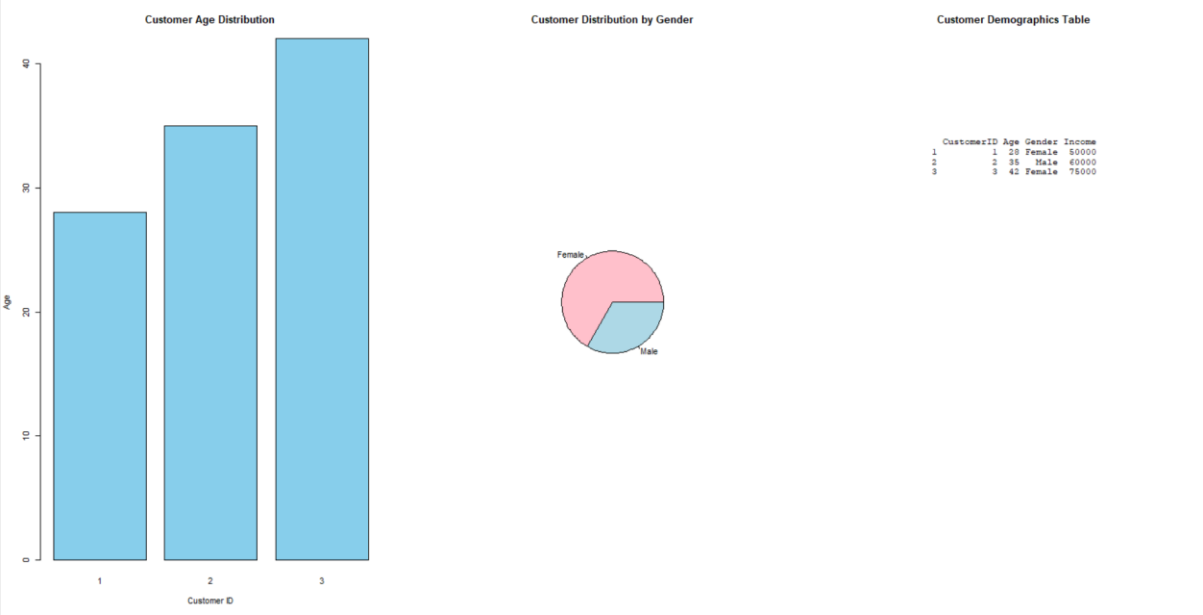
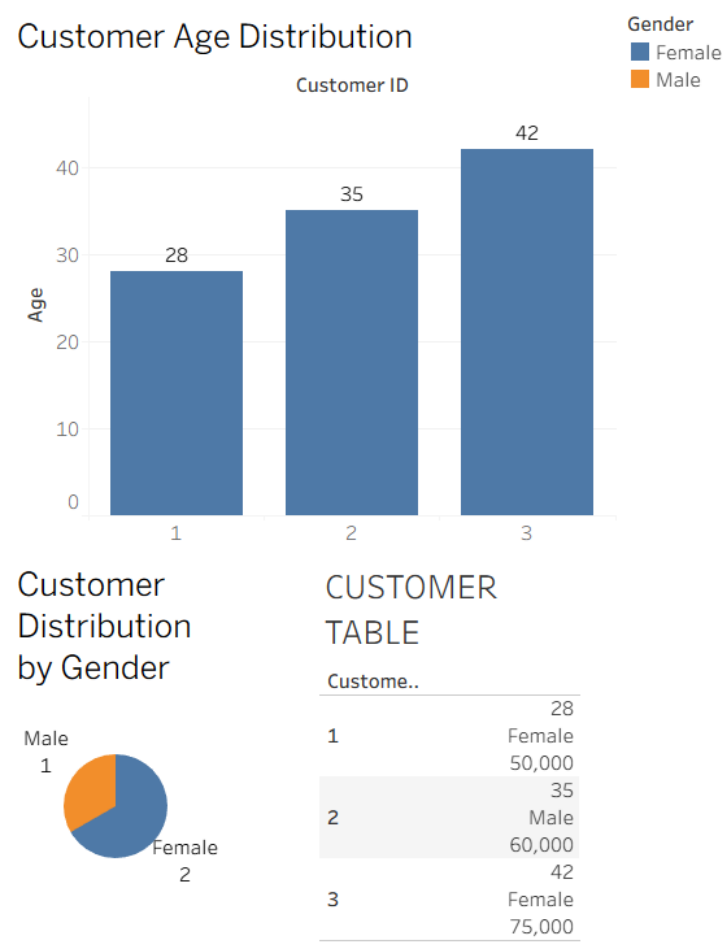


TABLEAU DASHBOARD:



Set 17: Employee Performance Analysis

R PROGRAM:

```
# Employee Performance Dataset
employee_id <- c(1,2,3)
department <- c("Sales","HR","Marketing")
years_service <- c(5,3,7)
performance <- c(85,92,78)

# Display 3 outputs together
par(mfrow=c(1,3))

# 1. Line Chart
plot(years_service, performance,
     type="o",
     pch=19,
     col="blue",
     main="Employee Performance Trend",
     xlab="Years of Service",
     ylab="Performance Score")

# 2. Bar Chart
dept_count <- table(department)

barplot(dept_count,
       col="lightgreen",
       main="Employees by Department",
       xlab="Department",
       ylab="Number of Employees")

# 3. Table
plot.new()
title("Employee Performance Table")

table_data <- data.frame(
  EmployeeID=employee_id,
  Department=department,
  YearsService=years_service,
  PerformanceScore=performance
)

text(0.5,0.8,
     paste(capture.output(print(table_data)),
           collapse="\n"),
     family="mono")

# Reset Layout
par(mfrow=c(1,1))
```

OUTPUT:

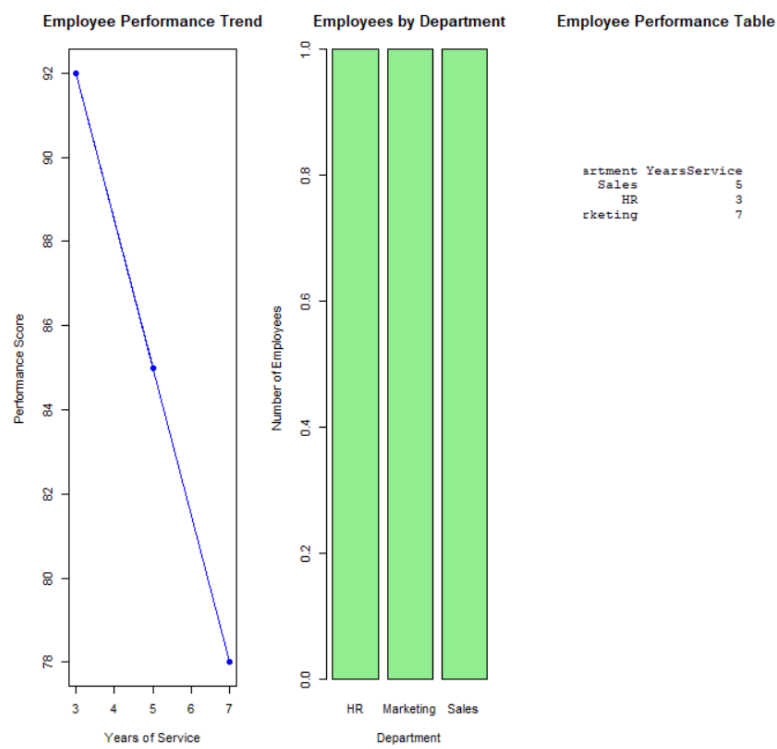
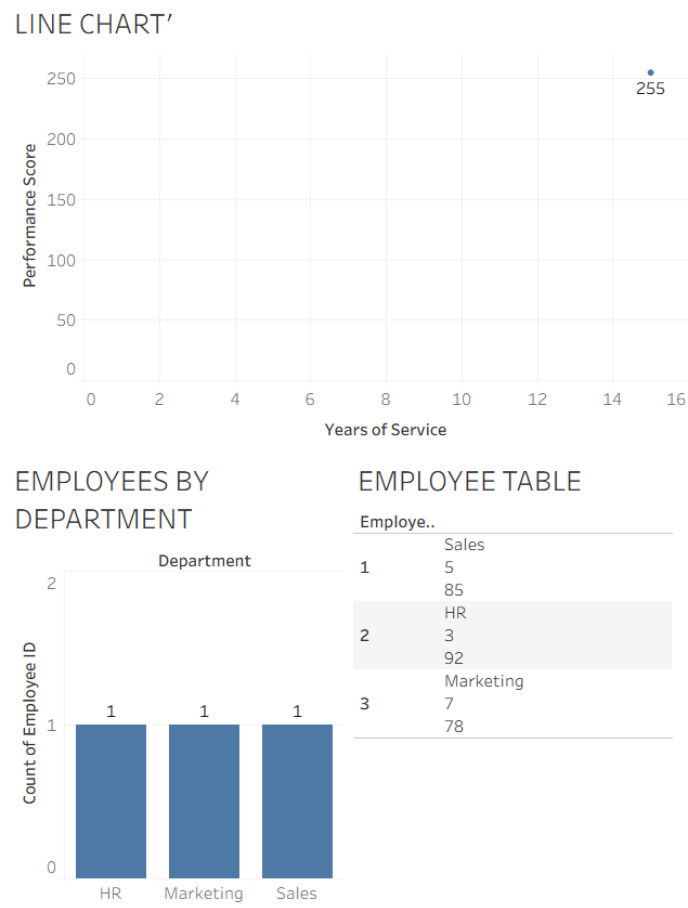


TABLEAU DASHBOARD:



Set 18: Product Inventory Management

R PROGRAM:

```
# Product Inventory Dataset
product <- c("Product A","Product B","Product C")
quantity <- c(250,175,300)
price <- c(20,15,18)

# Product Categories (Sample)
category <- c("Category 1","Category 2","Category 1")

# Display 3 outputs together
par(mfrow=c(1,3))

# 1. Bar Chart
barplot(quantity,
  names.arg=product,
  col="skyblue",
  main="Product Quantity",
  xlab="Products",
  ylab="Quantity Available")

# 2. Stacked Bar Chart
inventory <- table(category, product)

barplot(inventory,
  col=c("red","green"),
  main="Inventory by Category",
  xlab="Products",
  ylab="Count",
  legend.text=rownames(inventory))

# 3. Table
plot.new()
title("Product Inventory Table")

table_data <- data.frame(
  Product=product,
  Quantity=quantity,
  Price=price
)

text(0.5,0.8,
  paste(capture.output(print(table_data)),
    collapse="\n"),
  family="mono")

# Reset Layout
par(mfrow=c(1,1))
```

OUTPUT:

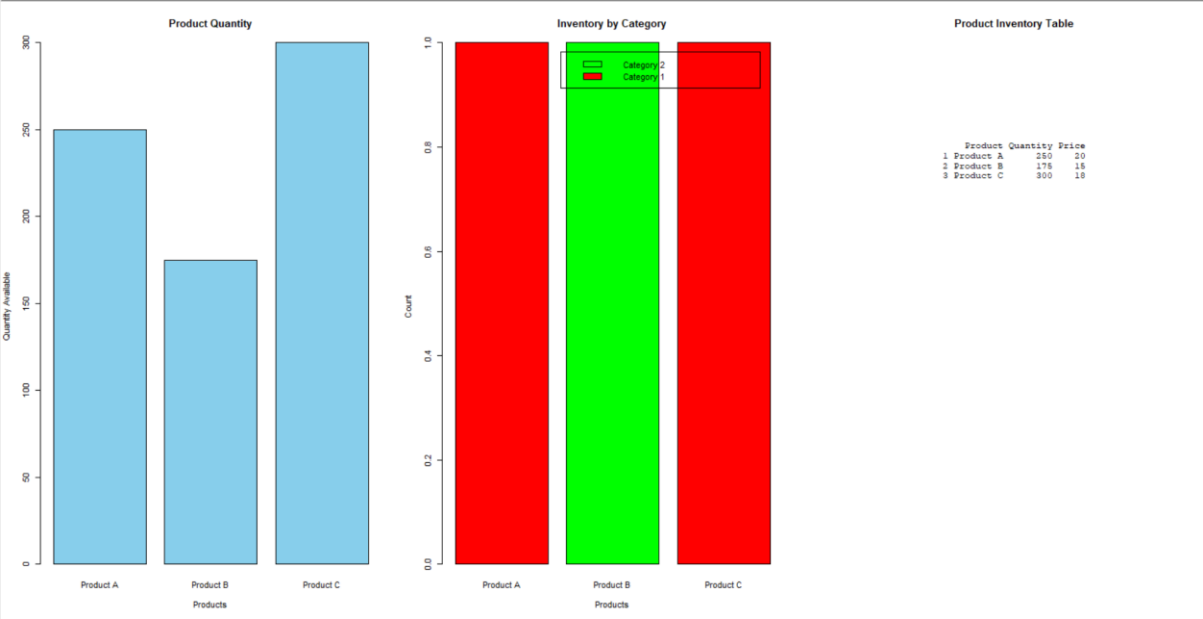


TABLEAU DASHBOARD:

